

BiclaNet – Reference Hardware Specifications

Conceptual system specifications for smart urban bike pod infrastructure

System Overview

Smart Pod → Cloud System → Operator Dashboard → Rider App

Core Components & Dimensions

Component	Dimensions (mm)
Unlocking System / Lock Controller	60 × 45 × 30
RFID / QR Access Reader	85 × 85 × 18
Bike Presence Sensor	18 × 18 × 8
Environmental Sensor Module	25 × 20 × 5
BiclaNet Hub (ESP32-class)	68 × 53 × 12
Internal Electronics Compartment (Envelope)	220 × 160 × 80

Functional Summary

- Secure physical bike locking
- Real-time occupancy detection
- RFID / QR secure rider authentication
- Cloud-connected control & monitoring

Connectivity & Data Flow

Sensors → BiclaNet Hub → Cloud → Dashboard & Rider App

Wi-Fi / LTE → Firebase / REST API

AI & ML Layer (high level)

- Usage prediction
- Occupancy forecasting
- Operator optimization insights

Safety & Fail-Safes

- Manual override
- Error logging
- Voltage & fault detection

“All specifications are conceptual reference values for Milestone 2 and may vary in final production.”